TECHNICAL DESIGN NOTE 02

Single-model inverse design from HFSS spectra

One neural network | fixed training configuration | no hyperparameter tuning | no inference-time optimization

HFSS CATALOGUE	MODEL INPUT	MODEL OUTPUT
160 standard spectra	**Band + absorption**	**Family + dimensions**

1. Exactly one trainable model

The system receives any band $450 \le a < b \le 1600$ nm and requested minimum absorption $0 \le X \le 1$. It outputs one of four shapes, full-cell dimensions from 0 to 320 nm and an air or SiO2 environment. There is one trainable network, no ensemble, no model comparison, no hyperparameter tuning and no geometry optimization at inference.



The one-network decision

Use a single multi-head residual MLP: one shared feature trunk, one classification output for the 8 geometry/substrate combinations, and one continuous output for the two valid dimensional coordinates. Multiple output heads belong to the same trainable model.

Input request	Output geometry
a: band start (nm)	One of 4 geometry families
b: band end (nm)	Air or SiO2
X: minimum absorption in [0,1]	Outer diameter or side (nm)
Operating condition: fixed TM, 0°	Inner diameter or side (rings only)

No-search specification: one predetermined architecture trained once. A deterministic offline reduction over available validated HFSS spectra generates supervision; there is no optimizer, candidate search or second network at inference. The model output is verified with a separate HFSS solve where possible.